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Private Industry Approaches to Network and Security Operations Centers

Best Practices for the Department of the Air Force



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About This Paper

How does private industry approach the challenge of ensuring the cybersecurity and performance of the computer networks they rely on? This paper describes how private companies commonly structure, organize, and staff network and security operations centers, highlighting best practices applicable to the Department of the Air Force's approach to protecting its own networks as it modernizes command and control. We also describe how advances in cybersecurity technology can reduce manpower requirements when properly implemented.

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Private Industry Approaches to Network and Security Operations Centers

The U.S. Department of War (DOW) began developing the concept of Combined Joint All Domain Command and Control (CJADC2) in 2019 as an effort to improve command and control by connecting systems across the land, air, sea, space, and cyberspace domains.¹ The Advanced Battle Management System (ABMS) division within the Department of the Air Force's (DAF's) Program Executive Office for Command, Control, Communications and Battle Management funds the development of architecture and systems engineering, digital infrastructure (DI), software and applications, and aerial networking required for the DAF's role in CJADC2. ABMS Digital Infrastructure (ABMS-DI) is a key component of this effort, developing a suite of fixed and deployable DI solutions that provide lightweight and scalable secure processing, connectivity, and data management for critical command and control functions. This new network infrastructure must be both efficient and resilient, requiring dedicated personnel and resources to manage the infrastructure, troubleshoot problems, and provide cybersecurity. The DAF has proposed standing up a command-and-control combined network and security operations center (C2NSOC) to support DAF operational elements using ABMS and ensure that ABMS performance and security meet the demands of CJADC2.

In contrast to the DAF, which focuses on mission objectives, private industry is looking to engage customers and maintain services that sustain revenue, and it depends on DI to do so. Organizations generally operate two core functions to maintain digital reliability and security: These are a Network Operations Center (NOC) and a Security Operations Center (SOC), respectively. This paper details common approaches used by NOC and SOC organizations in private industry, highlighting (1) which practices could improve efficiency in the DAF C2NSOC and (2) situations in which a different approach might be needed. We survey both NOC and SOC functions and responsibilities, organizational structures commonly used by private companies, and the corresponding manpower and staffing requirements. We also describe how advances in cybersecurity technology might reduce staffing requirements if properly implemented, and we conclude with lessons learned that the DAF could apply as the C2NSOC and ABMS-DI mature.

¹ The concept was originally *Joint All Domain Command and Control* (JADC2). DOW added the term *combined* in 2023. The Department of War is designated the Department of Defense under Public Law 81-216, National Security Act Amendments of 1949.

Functions and Responsibilities

The NOC ensures network availability and performance across on-premises, cloud, and remote environments. The SOC safeguards corporate assets from cyber threats that could disrupt operations, compromise data, or damage brand reputation.

Private-sector NOCs are primarily tasked with ensuring that services remain available to customers and employees. NOC analysts continuously monitor switches, routers, application servers, and cloud services for outages or performance degradation. Performance is usually measured in terms of network uptime, throughput, and latency, which directly affect availability of critical business services. Network disruptions can also expose the business to financial risk. When disruptions occur, NOC teams restore service through root-cause analysis and coordination with infrastructure and vendor support.

Cost-efficiency strongly shapes the private industry NOC. Staffing models often rely on lower-tier analysts supported by automation platforms; only problems that exceed routine troubleshooting are escalated to senior engineers. In efforts to reduce redundancy and optimize the use of potentially limited numbers of staff, many firms also consolidate NOC operations across business units or outsource them to managed service providers (MSPs) to reduce overhead while maintaining coverage 24 hours per day, 7 days a week (24x7).² In-house NOCs in private industry are therefore often as lean as possible, focusing tightly on availability and performance of business-critical systems.³

Private-industry SOC focus on protecting data, intellectual property, and customer trust. SOC analysts monitor logs and alerts from across the enterprise to detect intrusions, malware, and insider misuse. SOC performance is evaluated against detection times, incident resolution speed, and compliance outcomes. For firms in regulated industries—such as finance, health care, or retail—SOC operations also must demonstrate compliance with regulatory and industry requirements, such as the Payment Card Industry Data Security Standard, the Health Insurance Portability and Accountability Act of 1996 in the United States, and the General Data Protection Regulation in the European Union.⁴

Budget realities influence private-industry SOC operations similar to the ways those realities affect NOCs. Many companies adopt hybrid models in which routine alert monitoring is outsourced to managed security service providers (MSSPs), while in-house teams retain control of incident response, forensic investigations, and communication with executives. Private firms also are increasingly using automation platforms—such as security information and event management (SIEM) platforms and security orchestration, automation, and response (SOAR) tools—to compensate for staff shortages,⁵ although overreliance on tools and inadequate tuning can create blind spots in the security system and create opportunities for cyber threats.

The dividing line between NOC and SOC responsibilities might not be as clear cut in practice as we have outlined. The specifics of each center's operation (and corresponding size) are largely driven

² Suneel Wattal, "Maturity Model for IT Managed Services," *IOP Conference Series: Materials Science and Engineering*, Vol. 804, 2020.

³ Rick Blum, *Network Operations Centers*, Lucent Technologies, January 11, 2001.

⁴ Mark Orlando, *The State of Automation in Security Operations: A SANS Survey*, SANS Institute, June 2024.

⁵ Orlando, 2024.

by perceived risk. Increased operational risk, compliance requirements, and insurance obligations tend to increase both NOC and SOC size and responsibilities, which, conversely, decrease as a result of budget constraints and talent gaps. And, again, many private firms outsource some NOC and SOC functions, such as continuous monitoring, to MSPs or MSSPs while retaining in-house teams for incident response and strategic oversight.⁶ However, the NOC and SOC will both have some responsibilities that require close coordination with or without outsourcing.⁷ Both monitor enterprise systems using overlapping tools, such as log collectors, dashboards, and ticketing systems. A denial-of-service event illustrates the convergence: The NOC identifies traffic overload and reroutes flows while the SOC investigates whether the source is malicious and how to block it. Lack of coordination in such incidents risks extended downtime and increased financial loss.

Change management is another shared domain. Both NOC and SOC staff must ensure that when patches or configuration updates are deployed, these changes do not degrade service or open new vulnerabilities. Without adequate coordination, a regular configuration update that causes a temporary service outage might be interpreted by SOC personnel as a security incident.⁸

Reporting responsibilities also overlap, though the emphasis differs. The NOC provides leadership with metrics on uptime and service availability; the SOC reports on compliance posture and risk reduction. In private industry, these reports are not only operational documents but also instruments for demonstrating accountability to executives, regulators, and customers.⁹

Organizational Structures

NOC and SOC organizational structures in private industry can vary depending on company size, the amount of perceived risk to business operations from network performance issues and cyber threats, and available resources. Two structural dimensions shape how these centers are organized: centralized versus distributed deployment models and in-house versus outsourced staffing, as shown in Table 1.

⁶ Christopher Crowley and John Pescatore, *Common and Best Practices for Security Operations Centers: Results of the 2019 SOC Survey*, SANS Institute, July 2019.

⁷ Christopher Crowley, *SANS 2024 SOC Survey: Facing Top Challenges in Security Operations*, SANS Institute, May 2024.

⁸ Deepesh Shajee and Niles R. Ware, "Integrated Network and Security Operations Center: A Systematic Analysis," *IEEE Access*, Vol. 10, 2022.

⁹ Christopher Crowley, Barbara Filkins, and John Pescatore, *SANS 2023 SOC Survey*, SANS Institute, June 2023.

Table 1. NOC and SOC Structural Dimensions

		Location and Authority	
Resourcing		Centralized	Distributed
	In-house	Single NOC and/or SOC for entire company; staffed using internal personnel and resources	Multiple NOCs and/or SOC for different departments within company; each staffed using internal (to that organization) personnel and resources
	Outsourced	Single outsourced NOC/SOC for entire company	Multiple outsourced NOCs and/or SOC for different departments within company

Centralized Models Versus Distributed Deployment Models

The deployment model chosen reflects trade-offs among cost, efficiency, responsiveness, and control. A centralized model is one in which a single NOC or SOC handles operations for the entire firm. The NOC or SOC in this model usually operates from in a single physical location and might share space with other corporate information technology (IT) functions.¹⁰ This approach is common in industries in which economies of scale are important and network infrastructure can be monitored effectively from one location. According to the 2025 SANS SOC survey, the centralized model is still the most common, used in more than one-third (38 percent) of private firms.¹¹ Centralized operations reduce duplication of effort and simplify management oversight, but they also create potential single points of failure.¹² A localized disaster or system outage at the central facility could significantly hinder the ability to monitor and respond to incidents. Furthermore, a single-center approach can complicate responsiveness across global enterprises in which employees and customers span multiple time zones. Nonetheless, centralization remains attractive for many private firms because it is cost-efficient and provides the ability to maintain consistent procedures and toolsets across the enterprise.

A distributed model, in contrast, establishes multiple NOC or SOC facilities across regions or business units. This approach usually entails a higher cost to the organization but enables coverage across geographies and time zones, can more easily support local regulatory requirements,¹³ and enhances responsiveness in the event of a local facility outage. Distributed operations are common in multinational corporations with diverse business lines or strict regional compliance obligations, such as those in financial services and health care. However, distributed centers can also introduce coordination challenges and potential inefficiencies. For example, a distributed SOC or NOC in a midsized to large company can have less visibility into relevant data and wield less agency or authority

¹⁰ Since the coronavirus disease 2019 pandemic, more NOC and SOC organizations allow workers to be remote, mirroring the larger shift to remote work across multiple industries. However, the centralized model even in this case has staff located in one geographic region (Crowley, Filkins, and Pescatore, 2023).

¹¹ Christopher Crowley, *SANS SOC Survey 2025*, SANS Institute, July 2025.

¹² Abdoul K. Ganame, Julien Bourgeois, Renaud Bidou, and Francois Spies, "A Global Security Architecture for Intrusion Detection on Computer Networks," *Proceedings of the 21st International Parallel and Distributed Processing Symposium (IPDPS)*, IEEE Computer Society Technical Committee on Parallel Processing, March 26–30, 2007.

¹³ Crowley and Pescatore, 2019.

than a centralized version could. Having less coordination, authority, and access across the company in a breach response situation can create severe challenges to being able to remediate incidents. Maintaining consistent processes, alert triage standards, and communication across multiple centers requires significant investment in standardized tool integration and governance structures for coordination.¹⁴

One strategy that blends the distributed and centralized approaches is the *follow-the-sun* model, in which monitoring responsibilities pass sequentially among geographically distributed teams as global business hours progress. This approach ensures continuous coverage without requiring analysts in one location to work overnight shifts, and it is used by about one-half of the SOC's surveyed in 2024 that required 24x7 operations.¹⁵ Follow-the-sun structures are particularly valuable for multinational corporations and global service providers whose customers expect uninterrupted availability. This approach can also reduce analyst fatigue, a significant driver of turnover in SOC's, while still leveraging centralized oversight to maintain procedural consistency.

In-House Staffing Versus Outsourced Staffing

The distinction between in-house and outsourced staffing is particularly significant in private industry. In-house models provide direct control over sensitive data, allow tighter integration with internal business units, and facilitate rapid decisionmaking. However, the financial cost of maintaining a fully staffed NOC or SOC is prohibitive for many firms.¹⁶ Outsourcing can offer a less expensive option to rapidly gain critical network expertise, especially for smaller organizations.¹⁷ Outsourcing to MSPs (for NOC functions) and MSSPs (for SOC functions) has therefore become a common approach to handling functions that are specialized, repeatable, and resource-intensive.¹⁸ For example, in 2024, 22 percent of firms surveyed outsourced threat research, and 28 percent outsourced digital forensics.¹⁹ Outsourcing activities that require a skillset geared more toward going on the offensive is even more common; in 2024, 68 percent of firms surveyed outsourced penetration-testing functions and 58 percent outsourced Red-teaming functions.²⁰ Some MSSPs also offer continuous monitoring and triage functions for their clients, which can reduce costs and provide access to advanced expertise.²¹ Full outsourcing of all SOC capabilities can be done by employing cloud-based SOC services, which about 24 percent of SOC's surveyed in 2025 reported having done.²² However, outsourcing can increase the risk of misaligned priorities and delay incident response if contractual service-level agreements are not carefully defined. Private industry, as a result, often adopts a hybrid

¹⁴ Shajee and Ware, 2022.

¹⁵ Crowley, 2024.

¹⁶ Crowley and Pescatore, 2019.

¹⁷ Blum, 2001.

¹⁸ Crowley, 2025.

¹⁹ Crowley, 2024.

²⁰ Crowley, 2024.

²¹ Toby Bussa, Kelly Kavanagh, Pete Shoard, John Collins, Craig Lawson, and Mitchell Schneider, *Market Guide for Managed Detection and Response Services*, Gartner, August 26, 2020.

²² Crowley, 2025.

approach—keeping sensitive functions such as incident response, remediation, architecting, and compliance reporting in-house while outsourcing others.²³

Private industry’s preference for hybrid structures and selective outsourcing reflects the balance among operational efficiency, cost considerations, and flexibility in a dynamic threat environment. Smaller organizations might rely heavily on outsourcing to MSPs and MSSPs, while larger enterprises often retain core capabilities internally but supplement them with specific outsourced services and expertise. Regardless of the model chosen, private firms must manage the tension between cost reduction and operational effectiveness and must ensure that structural choices support both the availability of services and the security of data.

Staffing and Manpower Requirements

Private firms frequently address both cost and coverage demands by adopting tiered staffing models: Tier 1 analysts handle alert triage and escalation, Tier 2 staff conduct deeper investigations, and Tier 3 (or specialized roles) manage advanced incident response, digital forensics, or tuning of specialized tools.²⁴ Tables 2 and 3 describe common NOC and SOC roles, respectively. When NOC and SOC operations are integrated into a combined or *fusion center* model, overlapping responsibilities and unified monitoring systems can reduce redundant staffing positions. Cross-trained personnel capable of managing both performance and security telemetry enable a leaner workforce without compromising coverage. However, the combined environment often requires a higher average skill level per analyst, offsetting total headcount reductions with increased training and compensation costs.

Table 2. NOC Staffing Roles

Role	Skills
Tier 1 analyst or technician	Network-monitoring tools, basic device management, alert triage, ticketing systems
Tier 2 engineer	Advanced network troubleshooting, routing and switching, advanced industry certifications, scripting, Simple Network Management Protocol (SNMP), traffic analysis
Manager	Operations leadership, escalation handling, Service Level Agreement (SLA) enforcement, cross-team coordination
Network architect	High-level design, root-cause analysis, long-term planning, deep protocol knowledge

²³ Crowley, 2025.

²⁴ Shajee and Ware, 2022.

Table 3. SOC Staffing Roles

Role	Skills
Tier 1 analyst	SIEM operations, alert triage, log review, incident documentation, phishing analysis
Tier 2 analyst or incident responder	Malware analysis, endpoint triage tools, escalation, basic forensics, playbook execution
Tier 3 threat hunter or incident responder lead	Threat-hunting, threat intelligence, scripting, incident life-cycle proficiency
Manager	Incident response coordination, playbook tuning, reporting, compliance alignment, stakeholder communication
Security engineer	Tool tuning, log pipeline management, detection engineering, infrastructure hardening

Total NOC and SOC staffing levels in private industry are driven by organization size, operational coverage requirements, and functional complexity. SOC staffing requirements scale with the size of the organization, although there is a significant amount of variation.²⁵ In 2025, the most common size of a fully staffed SOC was two to ten people.²⁶

Table 4. SOC Team Analysts, by Organization Size

Organization Size	Common Number of Analysts
Fewer than 10,000	2–5
10,000–15,000	6–10
15,001–100,000	11–25
More than 100,000	26–100

SOURCE: Features information from Crowley and Pescatore, 2019.

Less information is available on NOC staffing size, though one survey found that one-half of NOCs in private industry employed fewer than ten full-time equivalent (FTE) staff, and another 22 percent employed ten to 24 FTE staff.²⁷ Other surveys have found that IT staff in general account for around 3 percent of employee headcounts.²⁸

The main consideration for both NOC and SOC staffing requirements is whether the organization requires continuous 24x7 support. Smaller firms with fewer than 1,000 employees typically operate NOC and SOC functions during business hours only, requiring a limited number of analysts who manage network availability and respond to security alerts as secondary duties. As

²⁵ Crowley, Filkins, and Pescatore, 2023.

²⁶ Crowley, 2025.

²⁷ Blum, 2001.

²⁸ Crowley and Pescatore, 2019.

organizations grow, however, service availability issues and security incident volume increase, necessitating a larger NOC and SOC even without continuous coverage. Many larger companies, especially those with a global footprint, opt for continuous 24x7 operational coverage,²⁹ and staffing requirements expand accordingly.

Some staffing models for a NOC or SOC that must operate on a continuous basis rely on a standardized multiplier to maintain uninterrupted monitoring and support, using an approach developed to estimate staffing for call center support. This approach calculates the number of hours per week required to staff a single position (such as a Tier 1 or Tier 2 analyst) 24x7 and adds an absence factor to account for vacation, illness, meetings, and other absences not related to work. Using an absence factor of 1.3, this means that an organization must allocate 5.46 FTEs for each workstation or analyst role requiring continuous staffing to sustain round-the-clock operations without service degradation.³⁰ This formulaic approach to staffing provides a consistent baseline but must be adjusted for operational complexity.

Other staffing models focus on maintaining an appropriate workload per analyst, using internal data on the average time that an analyst requires between opening and closing a service ticket and how frequently incidents occur.³¹ Higher numbers of staff per position can reduce the mean time to respond and maintain service level objectives. This approach can lead to more-accurate staffing estimates than the 24x7 multiplier but requires more data on frequency of incidents or service tickets and the time to resolve them.

In a NOC, operational complexity is driven by network topology, automation maturity, and the proportion of outsourced monitoring. Heavily automated environments that use advanced network analytics and self-healing mechanisms might reduce the number of Tier 1 analysts required, as described in more detail in the next section. Conversely, organizations managing legacy systems or complex network architectures often need additional engineers to handle manual troubleshooting and configuration management. The operational complexity for a SOC depends on the organization's threat exposure and regulatory requirements—that is, the perceived risk. Firms responsible for handling highly sensitive information (such as financial records, health care data, intellectual property, or government classified information) will likely want a more robust SOC staffing model to provide more redundancy and to reduce risk. SOCs that perform proactive threat-hunting, intelligence analysis, and Red- or Purple-team functions could also require additional FTEs beyond the baseline model.

Staffing for NOC and SOC operations in private industry ultimately must be treated as both a mathematical and strategic exercise. The 24x7 multiplier can provide a baseline manpower estimate, but the actual staffing requirement can be affected by multiple factors, such as use of automation, outsourcing, and organizational risk tolerance. Well-designed staffing models not only ensure operational continuity but also protect profitability and compliance posture, balancing the demands of constant vigilance with the realities of limited human capital.

²⁹ 79 percent of SOCs were operational 24x7 in 2025 (Crowley, 2025).

³⁰ This is derived from $1.3 * (24 \text{ hours per day} * 7 \text{ days per week}) / (8 \text{ hours per day} * 5 \text{ days per week}) = 5.46$.

³¹ Crowley, 2024.

Staffing Challenges

Staffing shortages present a major challenge to NOC and SOC organizations in private industry. Lack of experienced staff has historically been a significant challenge to improving NOC capabilities.³² Twenty-eight percent of survey respondents in 2024 named either high staffing requirements or a lack of skilled staff as their greatest challenge to the full use of their SOC capabilities.³³ Staffing shortages can be the result of several issues, such as the difficulty in training junior talent. Many of the skills required for higher-tier analysts in both NOCs and SOCs are more of an art than a science, requiring job experience and taking time to acquire. Furthermore, the lean posture taken by private industry leaves senior analysts with little time to step away from their core duties and train junior talent. Attracting and retaining already capable talent is also challenging; demand has historically outpaced the available talent pool. This demand makes it easier for high-performing NOC and SOC analysts to switch employers in pursuit of higher compensation, more-interesting work roles, and more-accommodating working conditions.

Technological Advancements' Impact on Manpower

Advancements in technology—specifically automation, integration, and artificial intelligence (AI) enhancement—have reshaped the staffing landscape of private industry, including the manpower requirements for NOC and SOC functions. The adoption of SIEM platforms, SOAR tools, and AI-enabled analytics has altered how operational coverage is maintained and how personnel are distributed across job roles. These technologies enable organizations to process vast volumes of data using fewer analysts, which increases efficiency and scalability, reduces errors, and saves costs.³⁴

Automated monitoring tools in NOCs can track network performance, identify anomalies in traffic behavior, and detect faults.³⁵ These tools allow network engineers to focus on remediation rather than detection. Automated configuration management tools and patching mechanisms can resolve minor outages without human intervention, thereby reducing the need for Tier 1 operators. These efficiencies do not eliminate the need for skilled personnel, however. Service outages caused by automatic patch updates show why NOCs must retain network engineers who can develop, maintain, and validate automated workflows, manage platform integration, and ensure that automated corrective actions do not introduce new faults.

SOCs using only manual processes struggle to detect and respond to incidents in a timely manner. Modern SOCs rely on endpoint detection and response systems and firewalls that use automated and AI-enabled tools to detect incidents. This approach helps solve the detection challenge but sharply increases the volume of security alerts that they need to address. SIEM platforms can leverage AI to filter, correlate, and prioritize alerts, increasing the efficiency of analysts and ensuring that they focus

³² Blum, 2001.

³³ Crowley, 2024.

³⁴ Despoina Giarimpampa, Roland Meier, Tegawendé F. Bissyande, Vincent Lenders, and Jacques Klein, "Exploring the Role of Artificial Intelligence in Enhancing Security Operations: A Systematic Review," *ACM Computing Surveys*, Vol. 58, No. 3, February 2026.

³⁵ Blum, 2001.

on the highest-priority incidents. Many SOC's also use SOAR platforms to manage cases, integrate threat intelligence, and automate responses through playbooks.³⁶ This technology can enable some response actions, such as host isolation or credential revocation, to occur in seconds rather than hours. This evolution in tools reduces the amount of time a SOC analyst has to spend doing redundant tasks, such as the continuous monitoring or alert triaging conducted by Tier 1 analysts.³⁷

Developments in AI and machine-learning over the past decade have the potential to further extend this transformation by assisting human decisionmaking.³⁸ Predictive analytics and anomaly detection models, when properly tuned, can identify deviations in network or user behavior that human analysts would likely overlook. For private industry, this capability translates into earlier detection of such potential incidents as insider threats or lateral movement by attackers. However, the challenges do not disappear; they instead shift the weight of the effort onto higher-skilled engineers who now must also understand AI models. These advanced tools also require regular retraining and validating using updated data on attack signatures to avoid bias or false positives.

Although these benefits bear consideration, overreliance on or the misuse of automation can pose significant risks for private firms. A commonly reported issue with automated detection tools is excessive alerts resulting from too many tools, redundant alerting mechanisms, or poor tuning; these alerts create more work for analysts rather than reducing their burden.³⁹ When tasks and processes are fully delegated to automated systems, human expertise can atrophy, leaving organizations unprepared to respond effectively when automated workflows fail. Automated response systems could also amplify errors when misconfigured or improperly tuned, potentially escalating benign anomalies into large-scale service disruptions. Excessive dependence on vendor-managed AI and SOAR tools can create visibility gaps and erode institutional knowledge, particularly when outsourced providers control the underlying models and data integrations.

Advances in cybersecurity technology thus pose a paradox for NOC and SOC organizations. Automation reduces the number of Tier 1 personnel needed for continuous monitoring, but it increases the need for technically proficient engineers capable of managing and validating those systems. The optimal manpower structure in highly automated environments might resemble an hourglass rather than a pyramid: a smaller base of Tier 1 operators, a narrow but highly skilled core of senior Tier 2 and 3 analysts, and a deep bench of engineers and architects responsible for the integration, tuning, and validation of automated systems.

The adoption of SIEM and SOAR platforms and advanced AI-enabled tools will not eliminate the requirement for human analysts in either NOCs or SOC's. Instead, this evolution should be viewed as a complement to human expertise to reduce repetitive tasks and augment the ability of humans to detect and respond to incidents faster and more effectively.

³⁶ Orlando, 2024.

³⁷ Kelley Dempsey, Arnold Johnson, Matthew Scholl, Kevin Stine, Alicia Clay Jones, Angela Orebaugh, Nirali Shah Chawla, and Ronald Johnston, *Information Security Continuous Monitoring (ISCM) for Federal Information Systems and Organizations*, National Institute of Standards and Technology, NIST SP 800-137, September 2011; Kathryn Knerler, Ingrid Parker, and Carson Zimmerman, *11 Strategies of a World-Class Cybersecurity Operations Center*, MITRE, 2022.

³⁸ Knerler, Parker, and Zimmerman, 2022.

³⁹ Orlando, 2024.

Conclusions

NOC and SOC organizations across private industry are designed to optimize performance while reducing cost. Best practices involve tiered staffing approaches that concentrate expert talent where it delivers the highest value. Operational efficiency is typically achieved through tight integration between cybersecurity technology and workforce planning. NOCs and SOC are structured to minimize redundant functions, using automated monitoring tools to consolidate alert streams and to route incidents based on severity and type. Regular assessments of both operations and tools are used to improve performance, tune automated systems to reduce false positives, and maintain alignment between network performance and security objectives. Many organizations leverage hybrid staffing models and outsource some functions to maintain worldwide operations without incurring excessive personnel costs. The result is a lean, metrics-driven operational environment in which every analyst's time is optimized.

However, direct transposition of some private-sector practices to DAF C2NSOC staffing requires significant caution. Private industry and the DAF share the technical challenges of continuous monitoring and incident response, but their operational priorities, risk calculus, and governance frameworks diverge sharply. Private firms operate under profit motives and optimize practices for efficiency, scalability, and compliance with regulatory standards. DAF operations, by contrast, are mission-driven, emphasizing network resilience and mission assurance over cost-efficiency. Practices that are optimal for a commercial enterprise—such as heavy outsourcing of NOC and SOC functions, aggressive automation of decisionmaking, or lean Tier 1 staffing—might introduce unacceptable risks in a military command-and-control context, such as the C2NSOC.

One critical difference lies in risk tolerance and operational authority. Private companies accept measured levels of downtime and incident exposure if the cost of full mitigation outweighs potential loss. DAF missions, by contrast, often operate under zero-tolerance conditions in which network degradation or delayed response can directly affect outcomes. Automation tools that make independent mitigation decisions without human validation, although efficient in industry, could conflict with the DAF's requirement for positive human control and chain-of-command accountability. Private industry will often outsource monitoring or incident response functions to external vendors as a way of reducing costs; this approach introduces security complexities for DAF networks that cannot rely on third-party access.

Personnel dynamics on the staffing side also differ substantially between the DAF and the private sector. Staff turnover in private industry is expected and offset through standardized training pipelines; direct hiring of needed knowledge, skills, and experience levels; and automation that captures institutional knowledge in workflows. Each organization has differences, but the functions that a NOC or SOC analyst is expected to perform and the skills they need are common across different private firms. DAF operations—and specifically C2NSOC functions—instead rely heavily on specialized, cleared personnel whose subject-matter expertise and skills are not easily replaced or automated.⁴⁰ Private industry can also reduce headcount more easily through increased use of

⁴⁰ Turnover is a significant challenge for DAF cyber-related operations, including C2NSOC functions. Permanent Change of Station rotations occur frequently, disrupting the ability of military members to achieve and maintain the subject-matter

automated tools and outsourcing. These approaches, along with integrating commercial AI decision engines, might not be applicable to the unique network environment that the C2NSOC must support.

Private industry offers valuable efficiency and scalability lessons for the DAF as it stands up the C2NSOC. The DAF should consider the following recommendations:

- Use tiered staffing models.
- Integrate workforce planning and cybersecurity technology acquisition.
- Leverage automated monitoring tools where possible.
- Perform regular assessments of operations and tools to improve performance.
- Use a follow-the-sun organizational structure to provide 24x7 coverage.
- Integrate NOC and SOC operations and structure.

However, these actions should not be applied wholesale to Air Force operations without adaptation. Rather, they must be recalibrated to emphasize mission assurance over cost reduction. A hybrid model that merges industry frameworks' focus on efficiency with the accountability and high performance required for DAF operations presents the most viable path forward for C2NSOC organization and staffing.

expertise required for highly technical positions. This can be mitigated by employing civilians and contractors but remains a major issue for the DAF.

Abbreviations

24x7	24 hours per day, 7 days a week
AI	artificial intelligence
C2NSOC	command-and-control combined network and security operations center
DAF	Department of the Air Force
FTE	full-time equivalent
IT	information technology
MSP	managed service provider
MSSP	managed security service provider
NOC	Network Operations Center
SIEM	security information and event management
SOAR	security orchestration, automation, and response
SOC	Security Operations Center

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